Arizona State University

Predictor-Compensated CLF–QP for Vehicle Tracking with Sensing and Actuation Delays.

Vikas Narang

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Agenda

- Background and Motivation.
- Problem Statement.
- Literature Review.
- Explanation of the Algorithm.
- Results and Discussion.
- Summary and Future Work.

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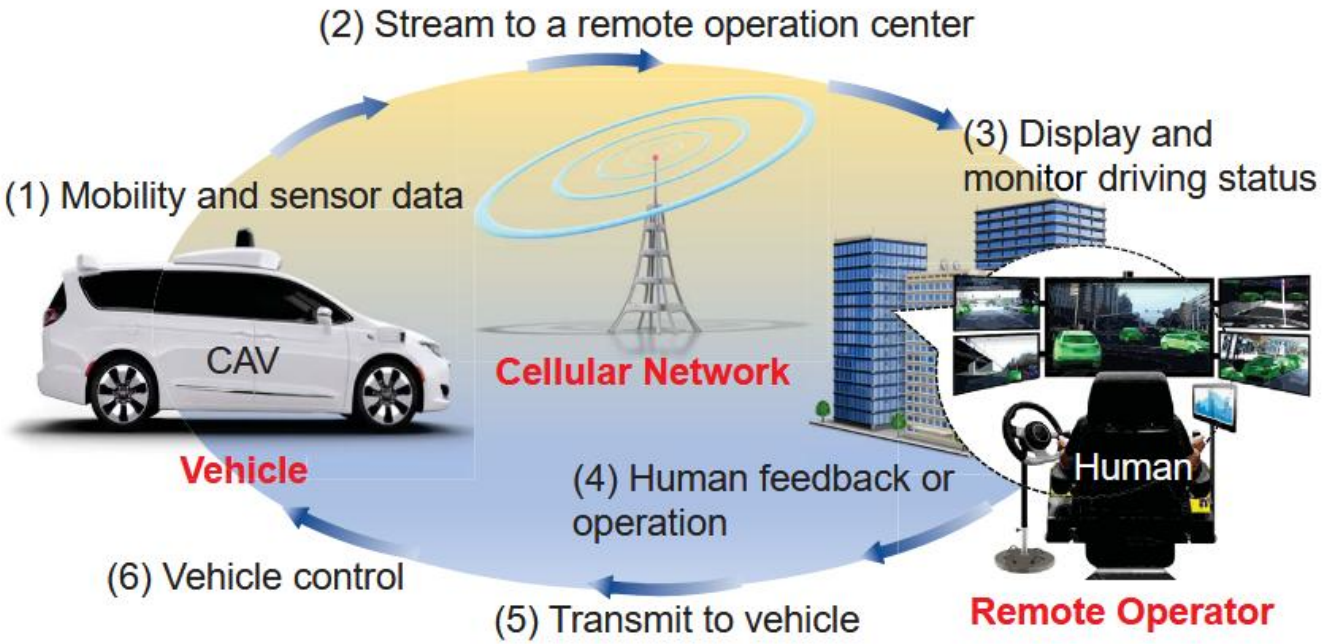
Introduction

- Teleportation refers to the remote control or supervision of a vehicle's motion by a human, using wireless communication, sensors, and actuators to perceive the environment and send commands without being physically inside.



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Background and Motivation

Background

- Modern control systems inevitably experience time delays from sensing pipelines, communication networks, computation, and actuator dynamics .
- In safety-critical applications such as autonomous driving, teleoperation, V2X communication, and multi-vehicle coordination, these delays can degrade tracking performance and even compromise stability.

Motivation

- Existing methods are limited when both measurement delays τ_1 and actuation delays τ_2 are present simultaneously.
- Predictor feedback ensures stability but does not directly enforce stability constraints.
- Delay-aware barrier approaches emphasize invariance but lack explicit performance guarantees and often treat only a single delay channel.

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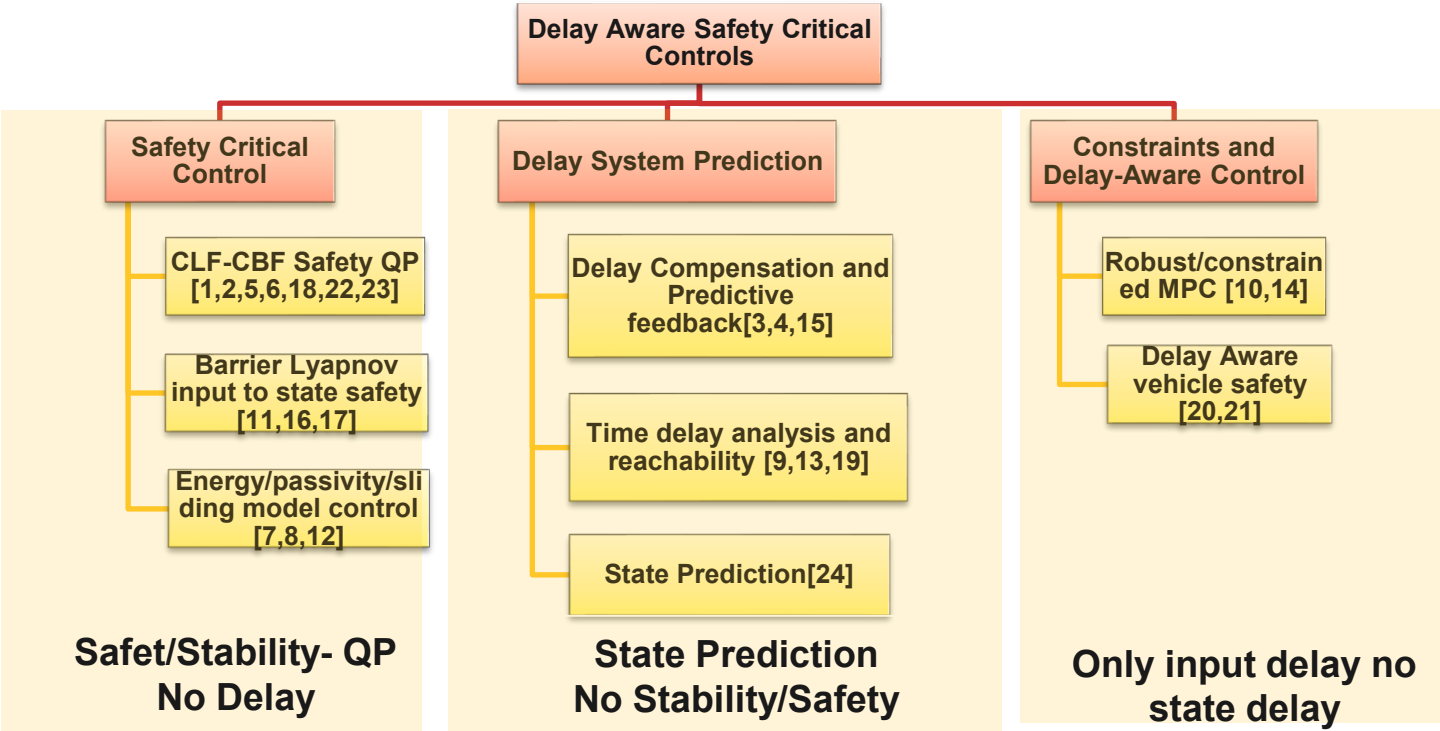
Tele-operation Literature Review



Research Gap

CLF-SQP framework that explicitly handles input/state delays, delivering formal safety and stability guarantees under bounded delays.

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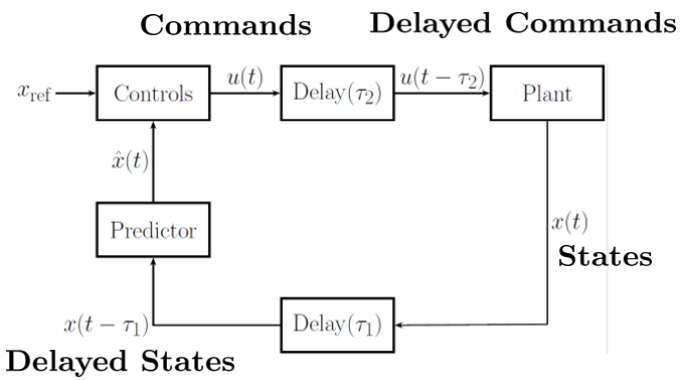
Problem Statement- System

Non-Linear Vehicle Kinematics Model

$$\dot{x} = v\ \cos\psi,$$
$$\dot{y} = v\ \sin\psi,$$
$$\dot{\psi} = \frac{v}{L}\ \tan\delta,$$
$$\dot{v} = a,$$



x and y are global coordinates; is ψ heading angle (rad); v is speed (m/s); a is longitudinal acceleration input m/s^2 ; δ is front-wheel steering angle (rad); and L is wheelbase (m).

Predictor

$$\dot{\hat{X}}(t) = f\Big(\hat{X}(t)\Big) + g\Big(\hat{X}(t)\Big)\ u(t-\tau_2) - g\Big(\hat{X}(t)\Big)\ K\big(\hat{X}(t-\tau_1) - z(t)\big).$$

Tracking error

$$e(t) = \hat{x}(t) - x_r(t), \qquad (Actual - Reference)$$

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Problem Statement- Optimization

Objective : Optimal actuation command using CLF-SQP under state and input delays.

CLF–QP

$$\min_{a_k,\, u_k,\, \Delta \geq 0} \quad \tfrac{1}{2}\left(R_{11} a_k^2 + R_{22} u_k^2 + q_\Delta\, \Delta^2\right)$$

$$\text{s.t.} \quad L_f V + L_g V \begin{bmatrix} a_k \\ u_k \end{bmatrix} + c\,V \;\leq\; \Delta,$$

$$a_{\min} \leq a_k \leq a_{\max},$$

$$u_{\min} \leq u_k \leq u_{\max}, \qquad u_k = \tan(\delta_k),$$

$$|a_k - a_{k-1}| \leq \dot{a}_{\max}\, T_s,$$

$$|u_k - u_{k-1}| \leq \dot{u}_{\max}\, T_s.$$

Why Δ and q_Δ:

$$L_f V + L_g V\, u + c\,V \;\leq\; \Delta$$

- $\Delta \geq 0$ is slack: it guarantees feasibility when constraints/rates prevent strict decrease.
- $q_\Delta\, \Delta^2$ penalizes slack use; larger q_Δ enforces the CLF more strictly.
- If feasible without relaxation, optimal $\Delta = 0$; otherwise $\Delta > 0$ absorbs violation at cost q_Δ.
- **PSD Hessian.** Cost $J = \frac{1}{2} z^\top H z + f^\top z$ with $H = \mathrm{blkdiag}(R, q_\Delta)$; if $R_{11}, R_{22}, q_\Delta > 0$ then $H \succeq 0$ (convex quadratic).
- **Affine CLF constraint.** $L_f V + L_g V\, z + cV \;\leq\; \Delta$ is linear in $z = [a_k, u_k, \Delta]^\top$.
- **Affine bounds.** Box/rate limits on a_k, u_k are linear inequalities; using $u_k = \tan(\delta_k)$ as the decision variable keeps them affine.

Variables Defination

- a_k: longitudinal acceleration command at step k (m/s^2).
- $u_k = \tan(\delta_k)$: steering command in u-space; δ_k is steer angle (rad).
- Δ: nonnegative slack for the CLF constraint.
- R_{11}, R_{22}: quadratic weights on a_k and u_k in the cost.
- q_Δ: quadratic weight on Δ.
- $a_{\min}, a_{\max}$: acceleration bounds.
- $u_{\min}, u_{\max}$: steering bounds (in u_k); correspond to $\delta_{\min}, \delta_{\max}$.
- $\dot{a}_{\max}$: acceleration rate limit (m/s^3).
- $\dot{u}_{\max}$: steering rate limit in u-space; corresponds to $\dot{\delta}_{\max}$.
- T_s: sample period.
- $V(\cdot)$: CLF; $L_f V, L_g V$: Lie derivatives.

$$V(\mathbf{e}) \;=\; \mathbf{e}^\top P\, \mathbf{e}, \qquad P \succ 0.$$

- c: CLF decay rate.

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Optimization Solver - SQP

- **Algorithm**
  - **fmincon** Algorithm= 'sqp'; each outer step solves a QP subproblem.
- **Decision Variables**
  - a_k, u_k, Δ .
- **Constraints and Bounds**:
  - bounds from amplitude and rate limits, rate limits referenced to previous command.
  - CLF: $L_f V + L_g V\, u + c\, V \;\leq\; \Delta$
  - No Equality constraints.
- **Objective:**
  - Quadratic with costs for a_k, u_k, Δ .
- Warm start: Initialize with previous solution z_{k-1} to reduce iterations.

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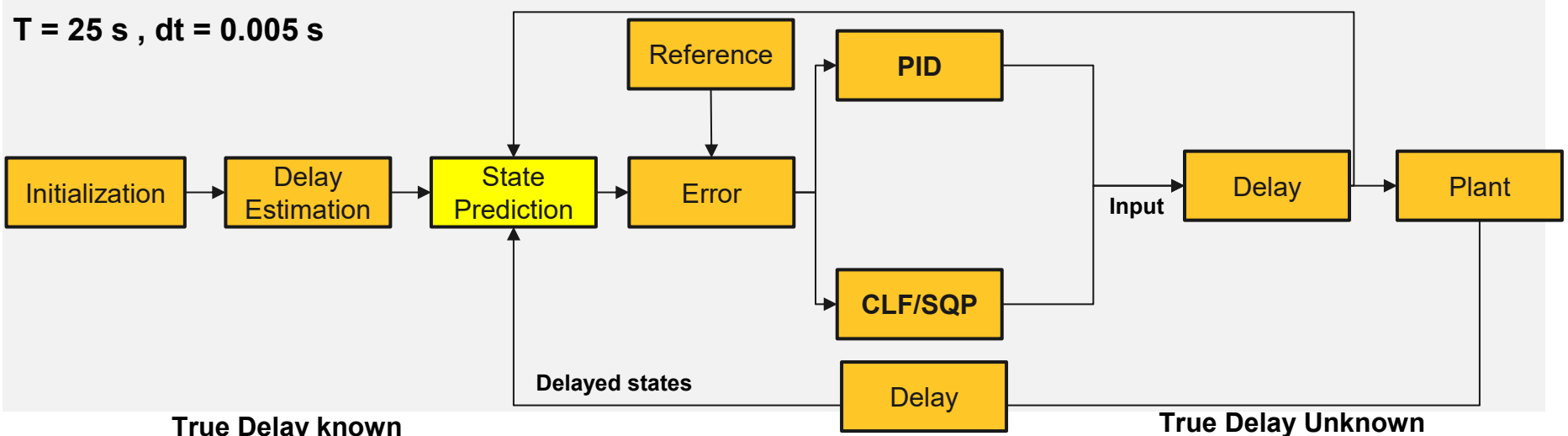
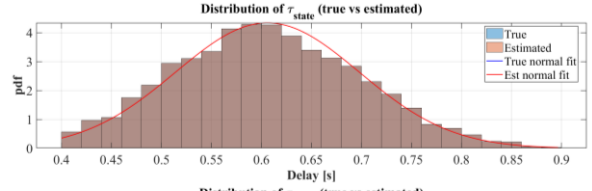
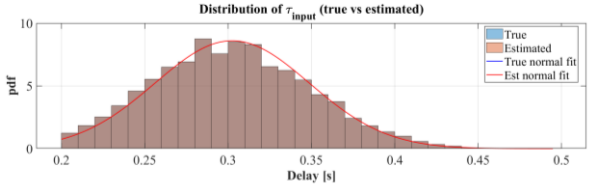
Simulation Setup

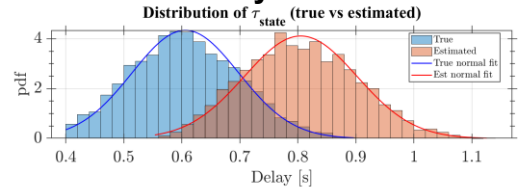


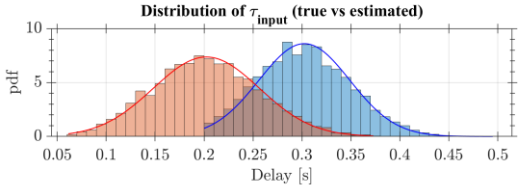
True Delay known

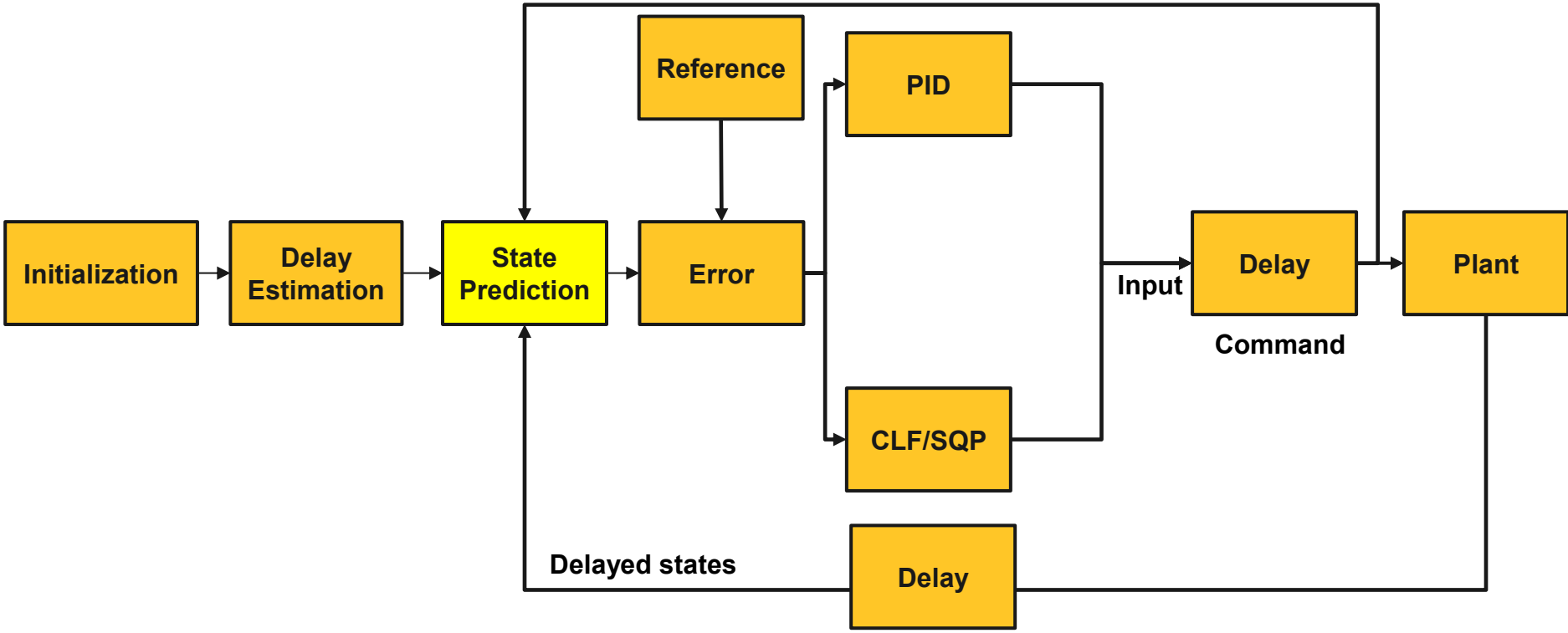




True Delay Unknown







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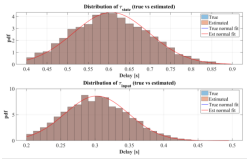
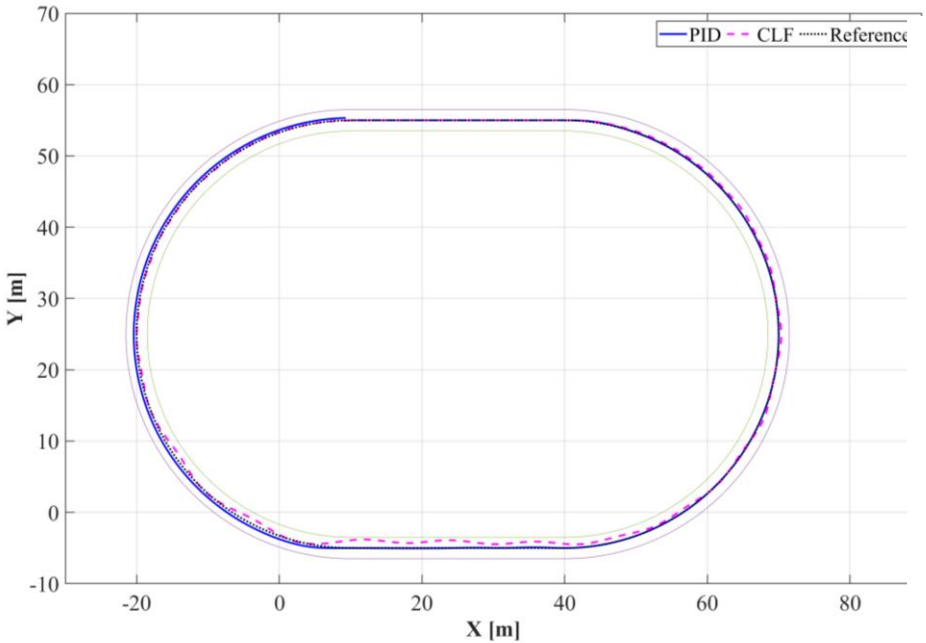
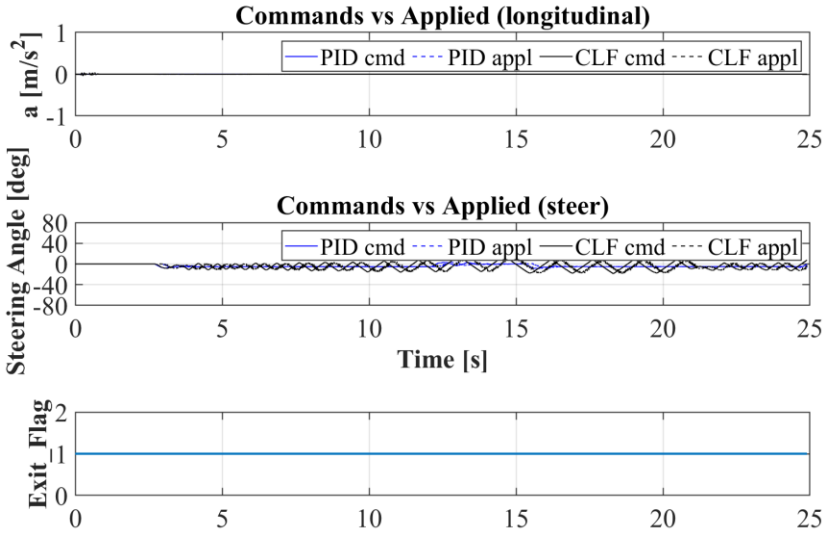
True Delay known



Results –True Delay known





Controller	RMSE [x(m)]	MaxAbs [x(m)]	RMSE [y(m)]	MaxAbs [y(m)]	RMSE [psi(rad)]	MaxAbs [psi(rad)]	RMSE [velocity(mps)]	MaxAbs [velocity(mps)]
PID	0.44	1.20	0.23	0.44	0.01	0.09	0.00	0.00
CLF - SQP	0.30	0.67	0.40	1.09	0.08	0.19	0.00	0.00

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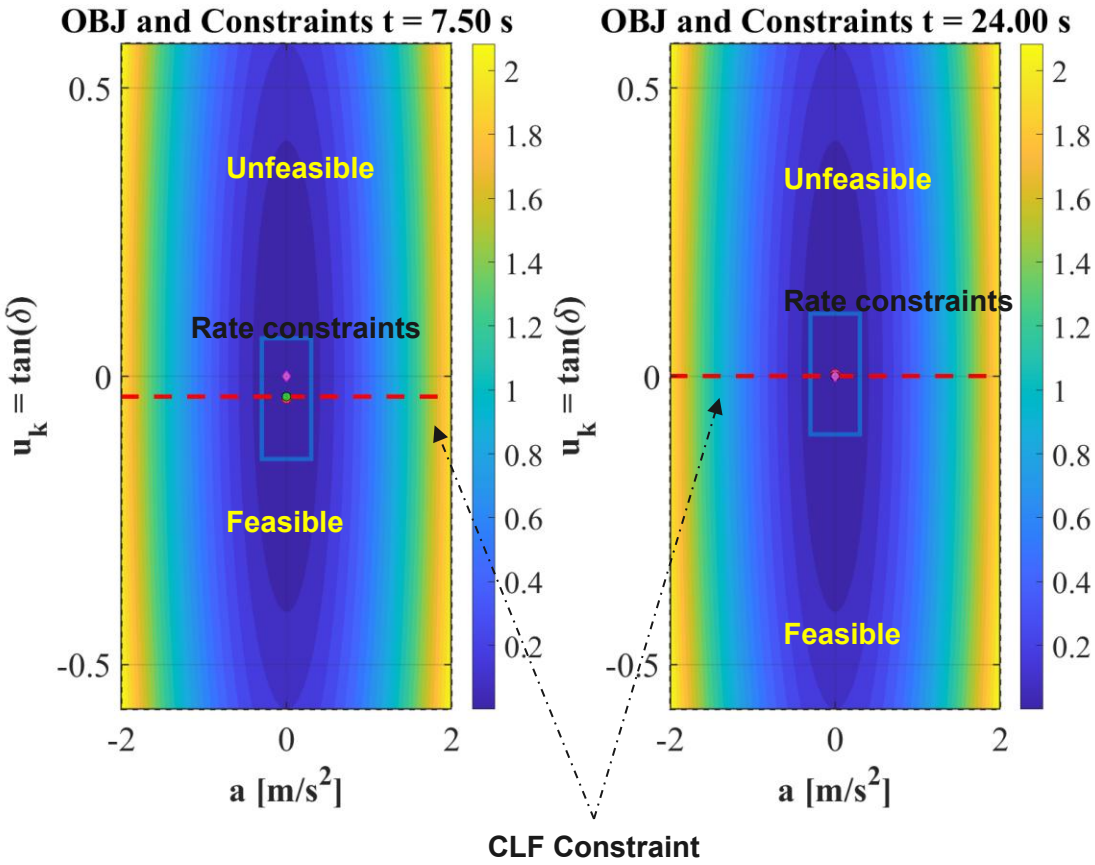
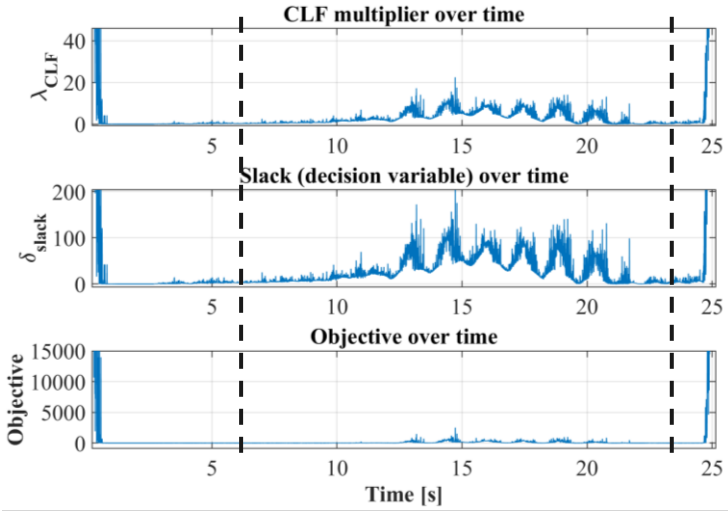
Results –True Delay known





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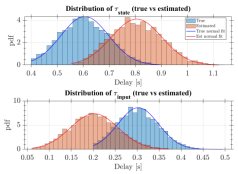
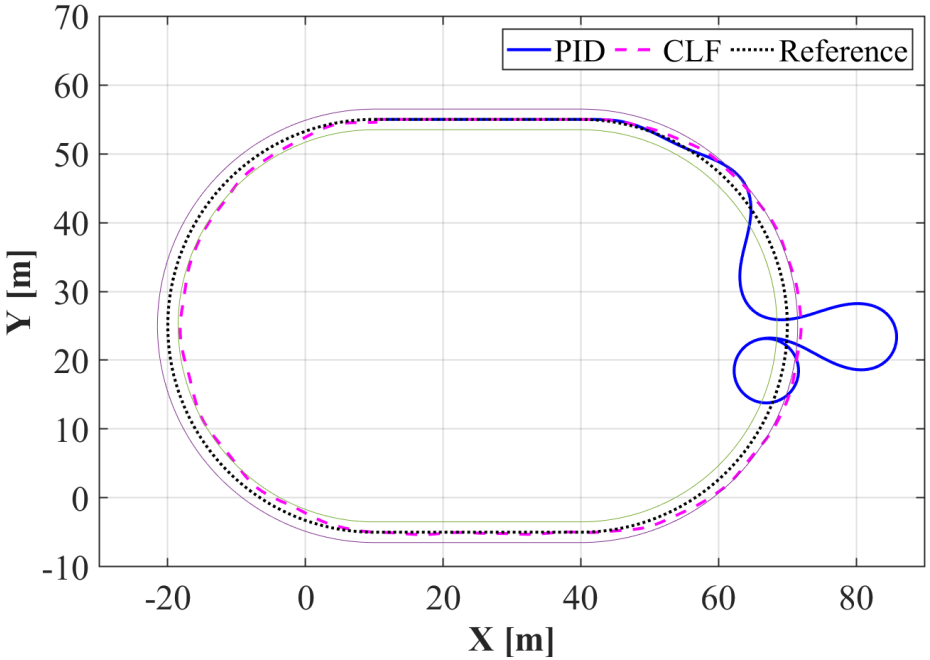
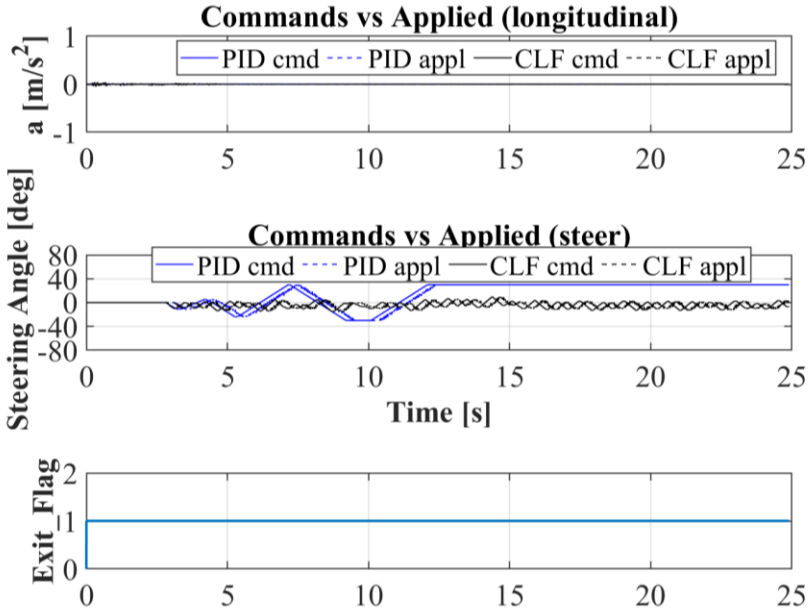
Results –True Delay Unknown

True Delay Unknown







Controller	RMSE [x(m)]	MaxAbs [x(m)]	RMSE [y(m)]	MaxAbs [y(m)]	RMSE [psi(rad)]	MaxAbs [psi(rad)]	RMSE [velocity(mps)]	MaxAbs [velocity(mps)]
PID	31.1	57.4	43.3	91.8	1.4	3.1	0.00	0.00
CLF - SQP	2.4	3.8	1.1	2.2	0.07	0.1	0.00	0.00

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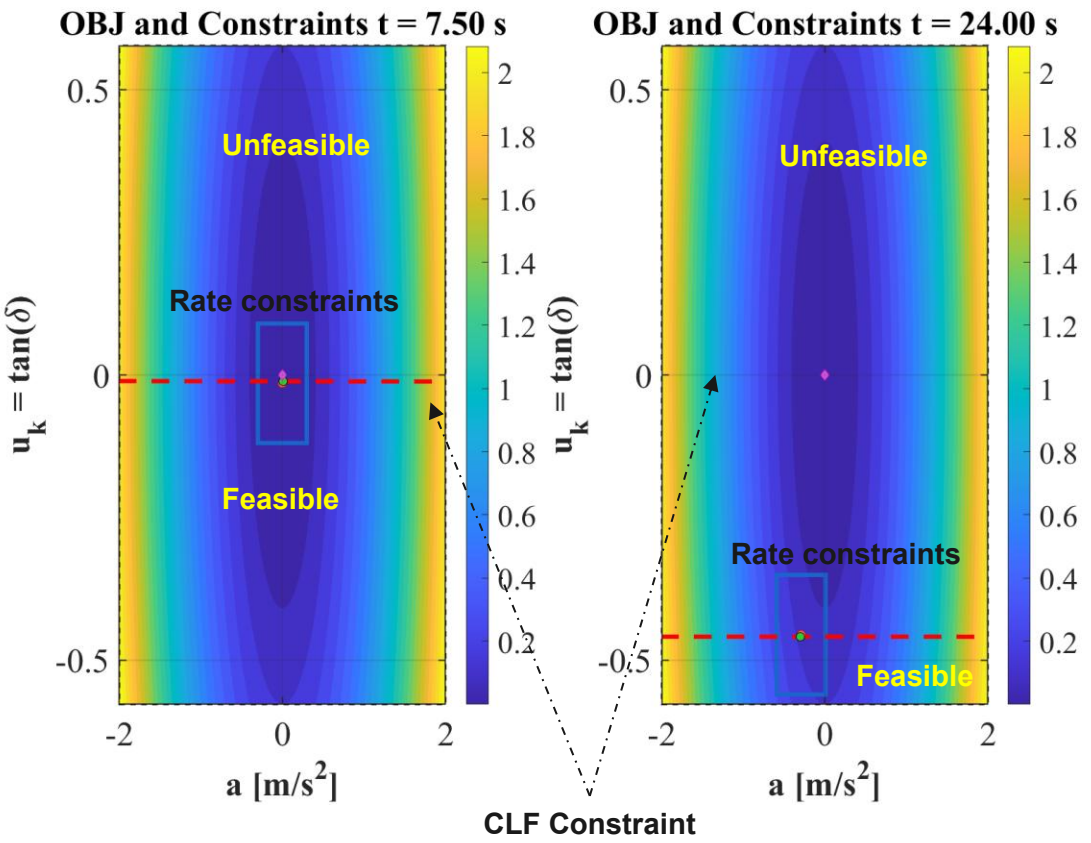
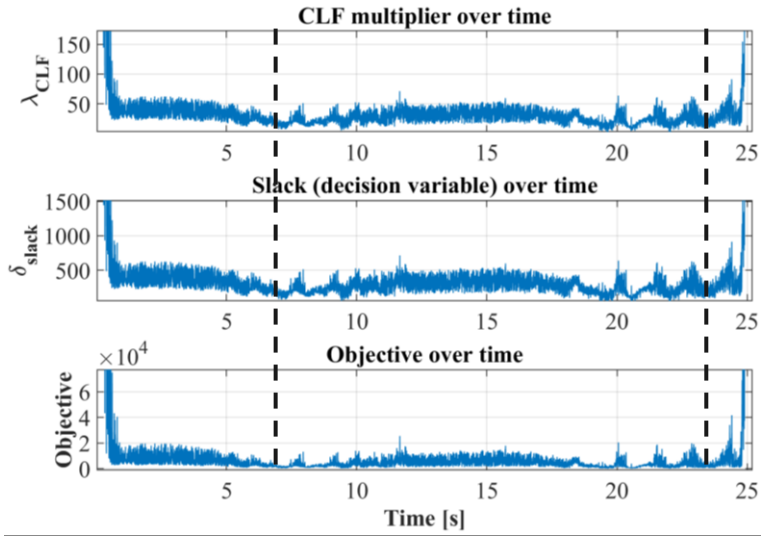
Results –True Delay Unknown





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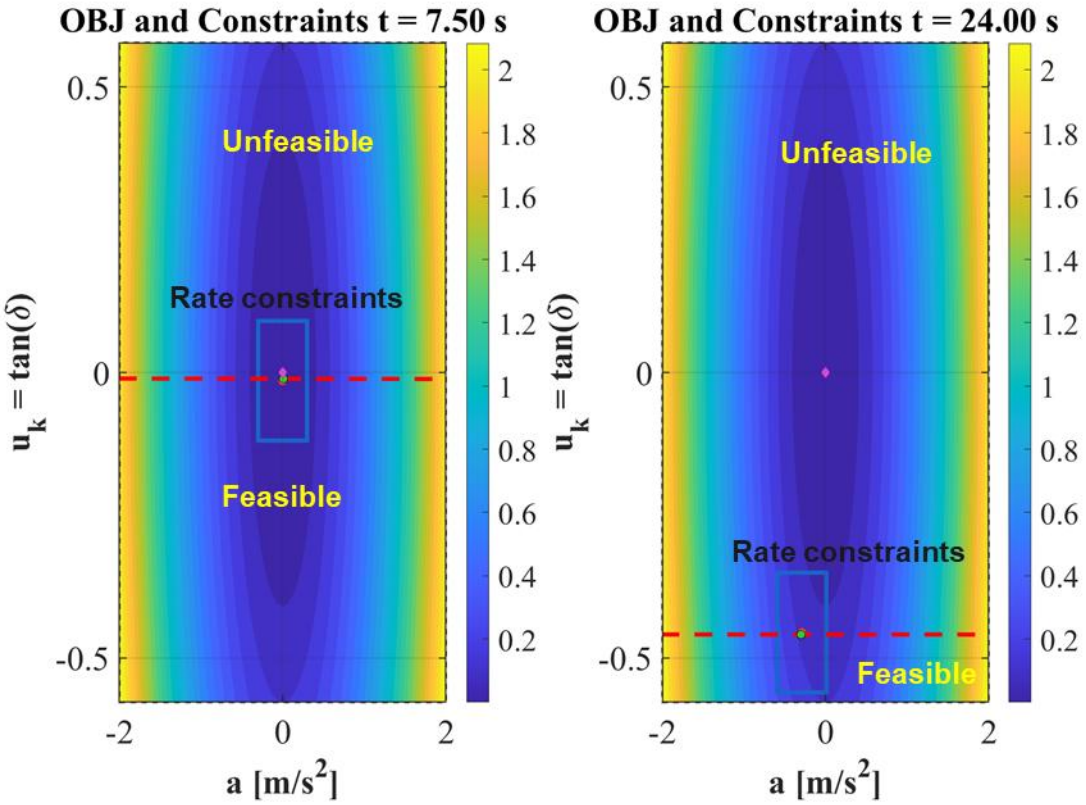

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Conclusion

- **Objective and Guarantees.** The CLF-QP enforces $\dot{V} + cV \leq \Delta$ directly encoding exponential decay of the Lyapunov function. PID lacks a formal stability certificate under delay and relies on tuned gains plus prediction.

- **Constraint handling** CLF-QP respects amplitude and rate limits via box constraints in the QP; PID saturates after the fact, which can prolong error transients when saturation is active.

- **Delay handling** Both use the same predictor/preview; however, CLF-QP solves for commands consistent with the predicted arrival state and Lyapunov decrease, making it more resilient to bias in delay estimation.

- **Computation** The SQP-based QP is small (three variables with one CLF inequality and box/rate bounds). The logged exit flags and objective plots indicate reliable convergence at 200 Hz.

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Summary of Learning/Future Work

Summary of Learning

- Lyapunov-centric objective: Encoding ($\dot{V} + cV \leq \Delta$) in the QP yields formal stability decay, unlike PID tuning; slack (q_Δ) sets the rigor of enforcement.
- Convexity and solvability: With PSD/PD weights and affine CLF/CBF, box, and rate constraints, the problem remains a small convex QP (3 vars, one inequality), giving reliable convergence.
- Slack trade-off: (Δ) preserves feasibility under competing constraints; tuning (q_Δ) balances strict Lyapunov decrease.

Future Work

- Develop an optimization-based controller that simultaneously enforces safety (CBF) and performance (CLF), with both constraints active in the same solver.
- Model uncertainty: represent dynamics/modeling error (e.g., bounded disturbance).
- Extend from the current kinematic bicycle model to nonlinear dynamics and adapt the predictor plus CLF/CBF constraints to the richer model.

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Thank you !

Questions?

Name : Vikas Narang | Email : vnarang2@asu.edu

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